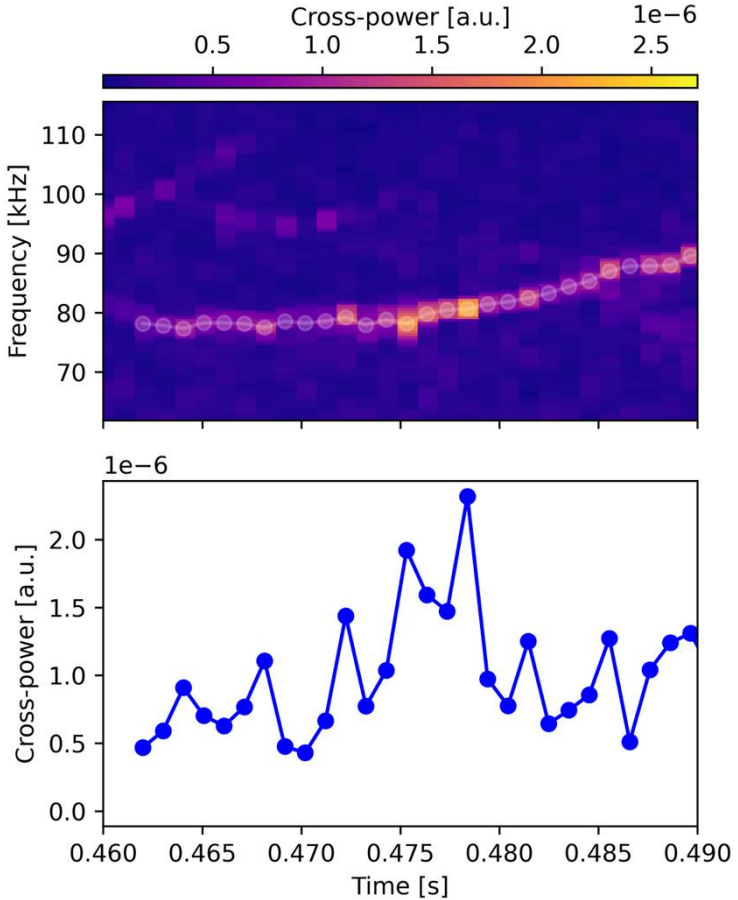
How to track fluctuation with physics-based algorithms

by

Jose Rueda-Rueda[1] & P. Oyola[2]

1) Department of Physics and Astronomy. University of California Irvine
2) Princeton Plasma Physic Laboratory



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UCI PPPL

The amount of data produced in a run day can't be analyzed by simple human search

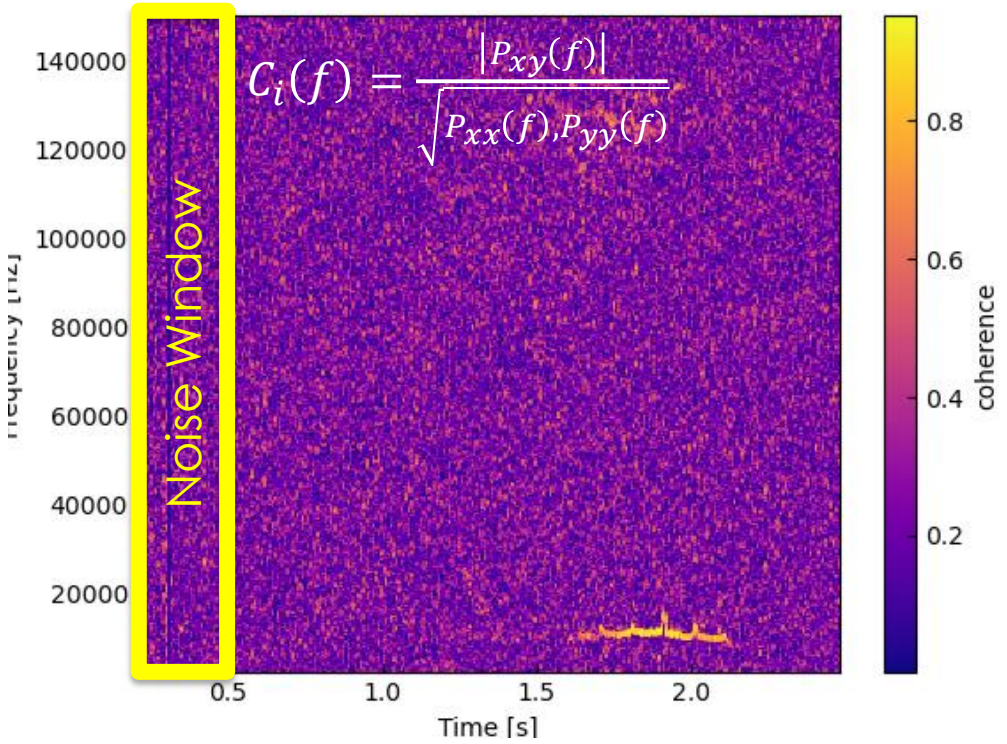
- EP fluctuation diagnostics does not produce outstanding modes in the autopower -> coherence with other fluctuation diagnostics must be analyzed
- The minimum amount of spectrum to analyze would be:
  - 24 EP ch x (32 BES + 20 ECE + 6 MAG) x 15 shots > 20 000!!
- Adding also other measurements in piggyback, this number grows over a hundred thousands
- It is impossible to revise each combination manually

Outline

- **A fast filter to find modes**
- **A *Travelling Salesman* Algorithm to track modes**

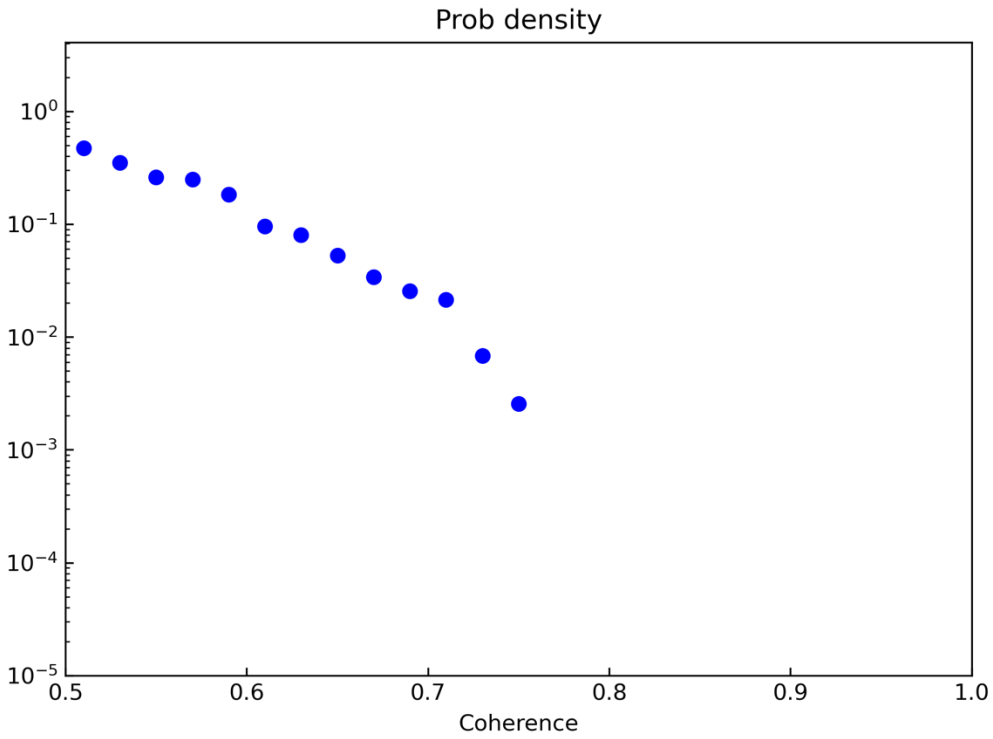
~150 ms time window needed for good statistics for fit

- Collect all bins in the noise window and fit to a model



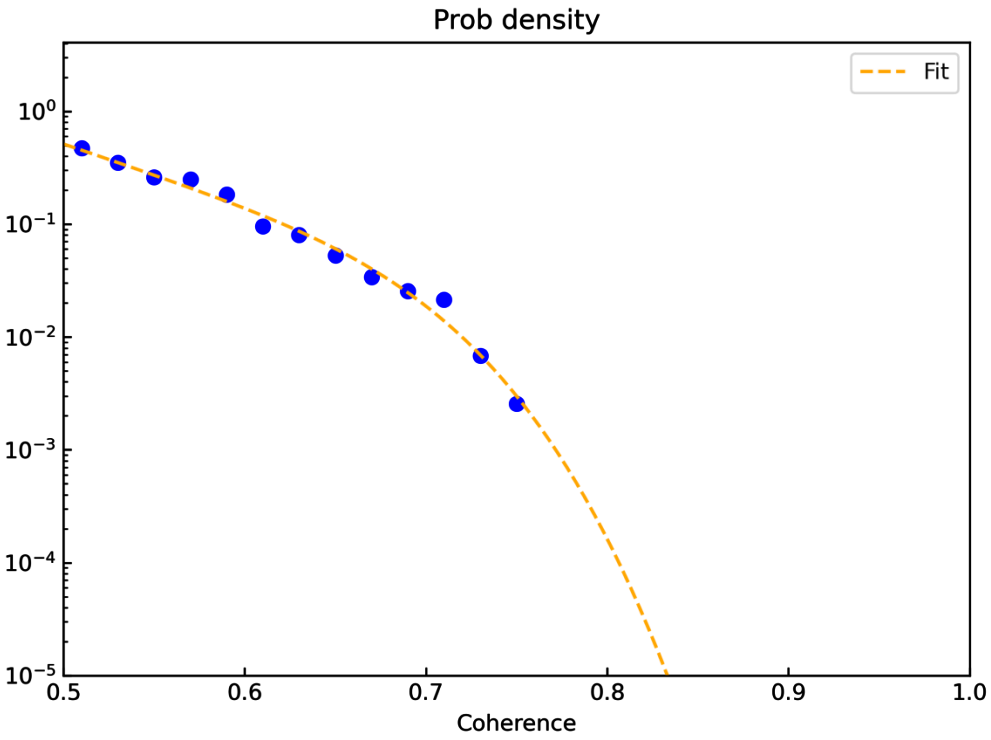
1st Get all the noise in the window

- Collect all bins in the noise window and fit to a model
- Definition of noise window:
  - Somewhere where modes are not expected (ie 0.2-0.4s)



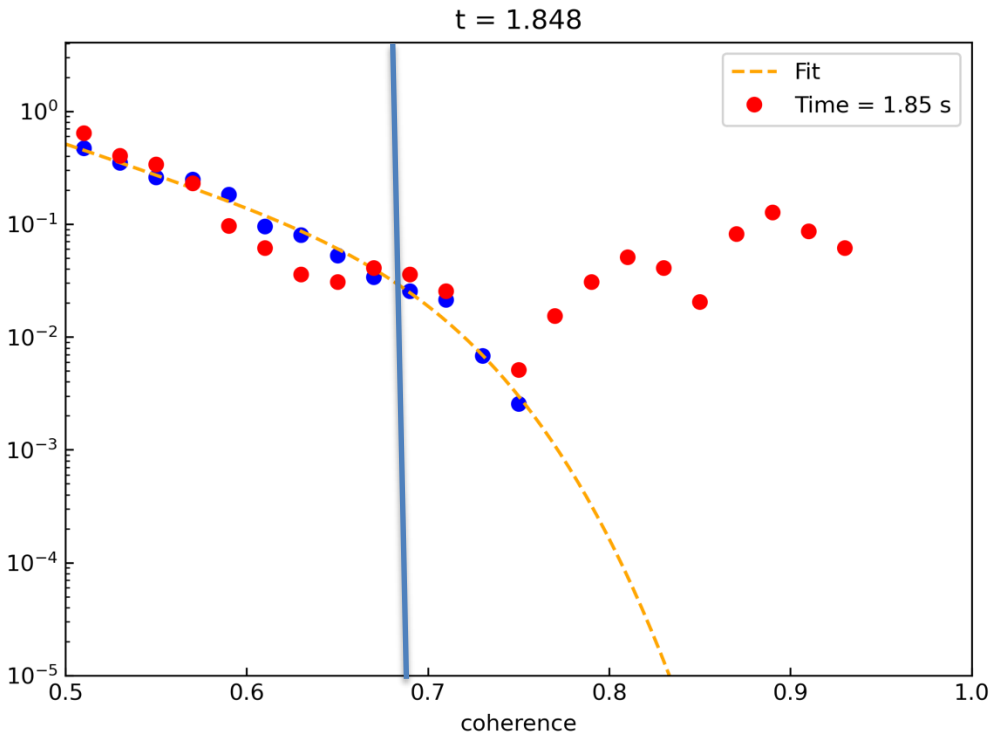
2nd Fit the noise histogram to a polynomial

- A simple 2nd or 4th order polynomial gives good results ~$r^2 = 0.94$



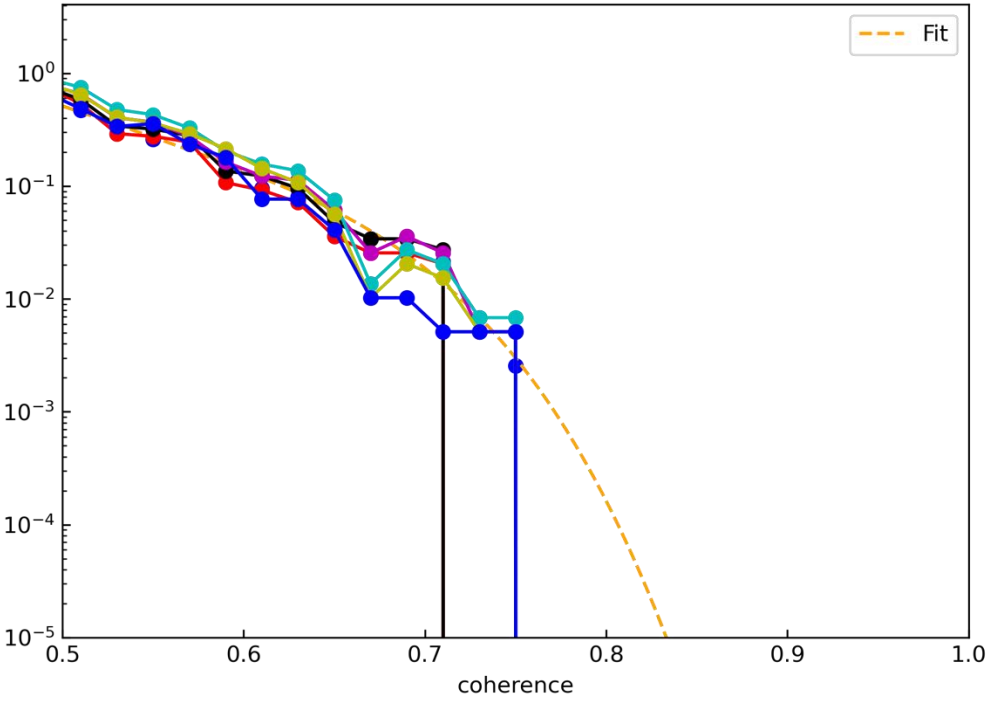
3rd Compare with the time slice you want

- Compare the fit to your time of interested
  - $Accept = \left(\xi \equiv \frac{\int_{\{x>x_0\}} H(t_i) - H_{fit} dx}{\int_{\{x>x_0\}} H_{fit} dx} > 0 \right)$
- This will get false positives



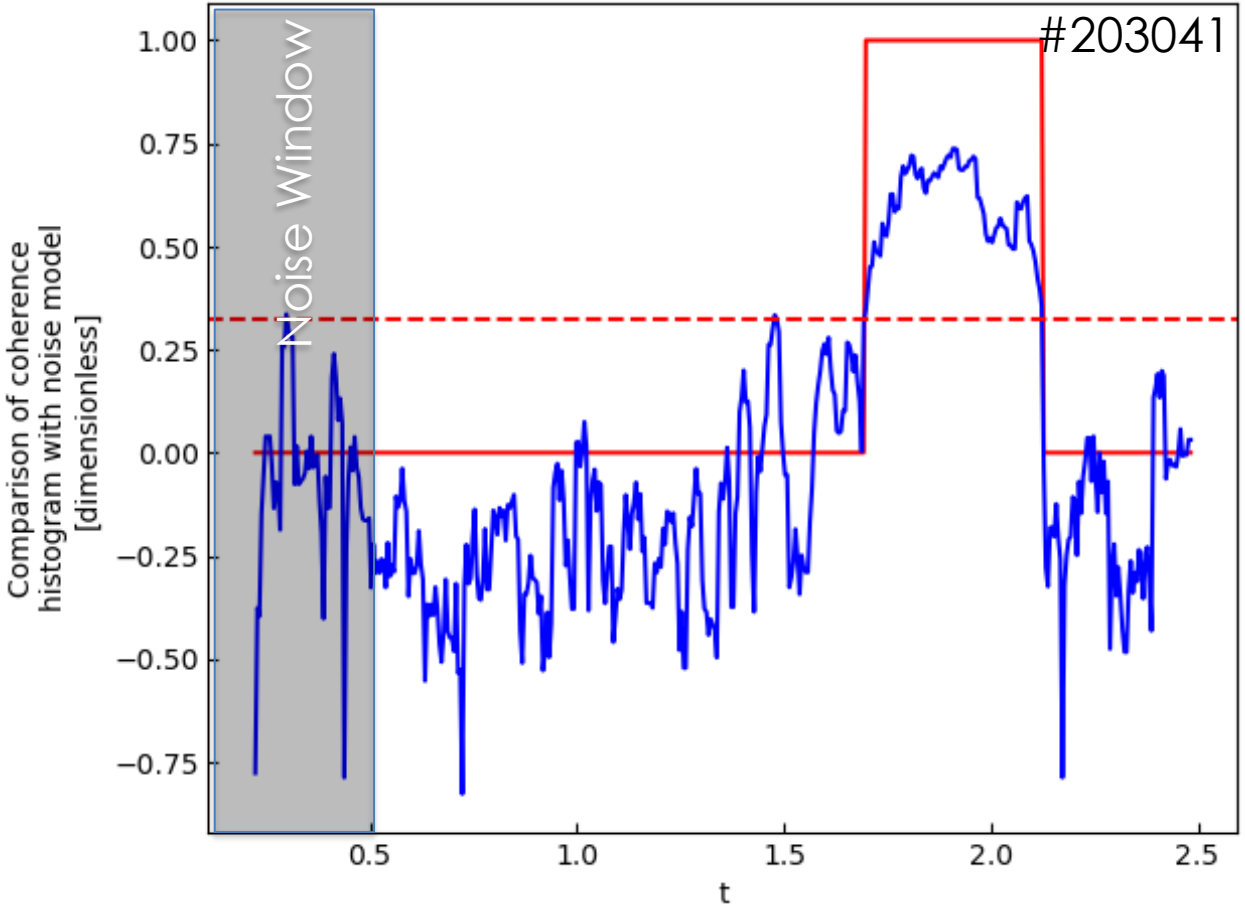
Statistical noise require a different threshold to accept a mode

- Natural fluctuations makes the coherence not to match the fit when single time slices are chosen
- Solution: Collect all time slices in the noise window, compare with the fit an define a mean and dispersion in the comparison with the mean

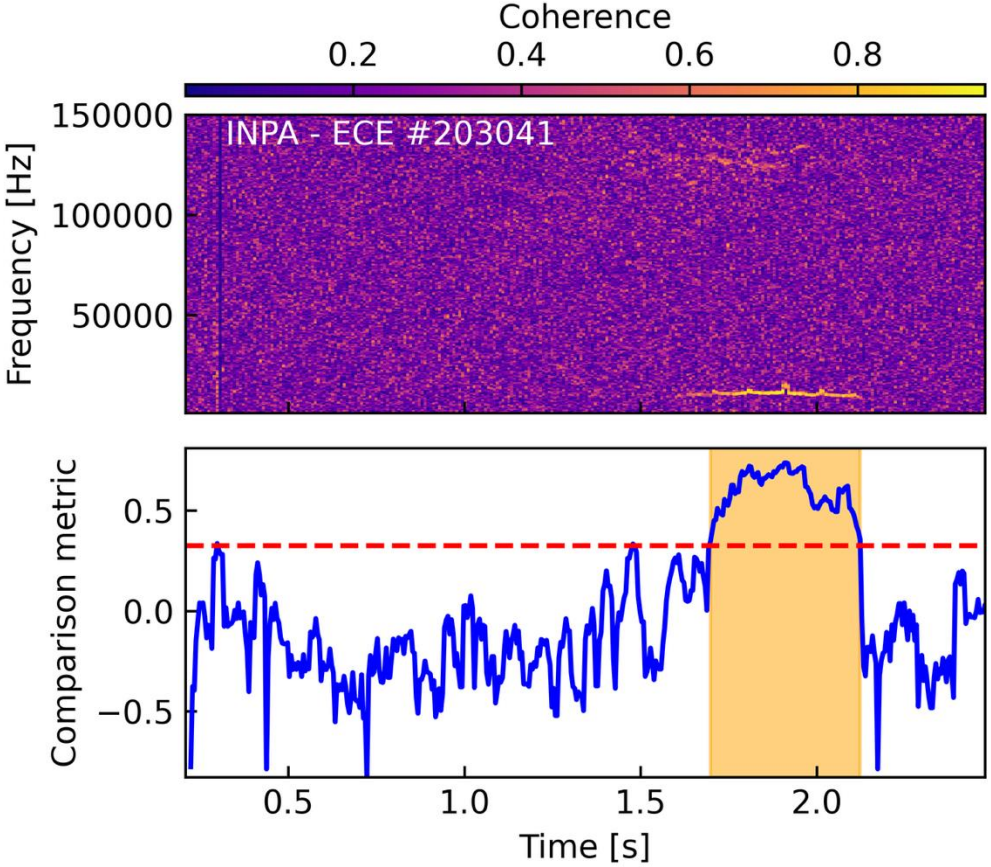


Simple comparison to mean and standard deviation gives the mode location in time and the likelyhood of being a mode

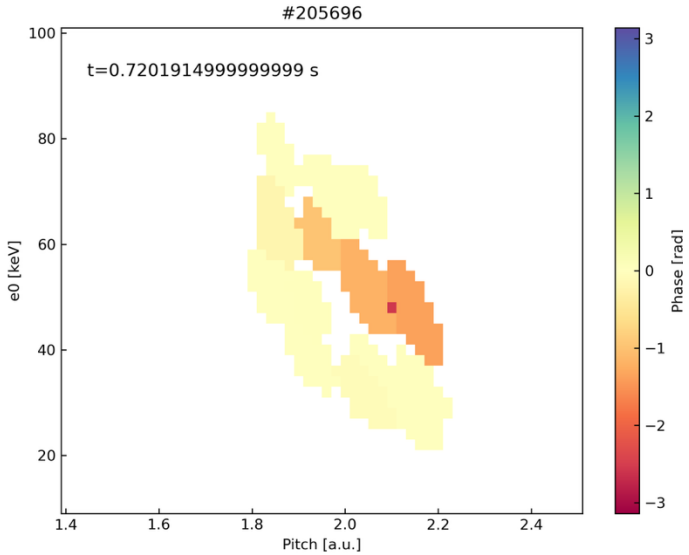
- Model parameters:
  - Confidence threshold
  - Window size (if cumulating several time slices)
- Extra condition:
  - Discard cases where positive comparison are shorter than a given time length

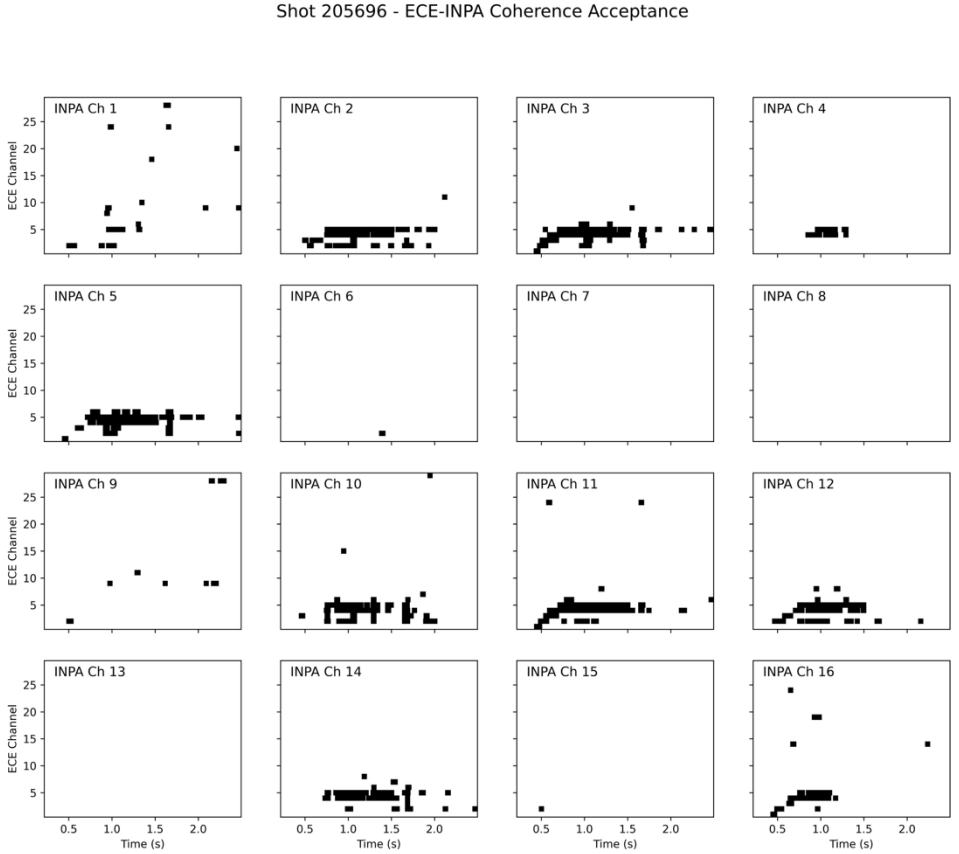


Time range predicted by the comparison match the actual region where the mode is present



#205696 as an example of the algorithm working



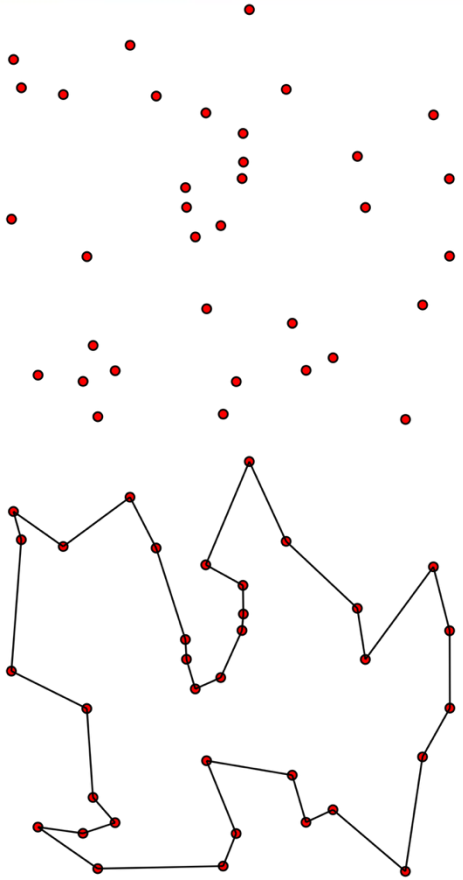


Outline

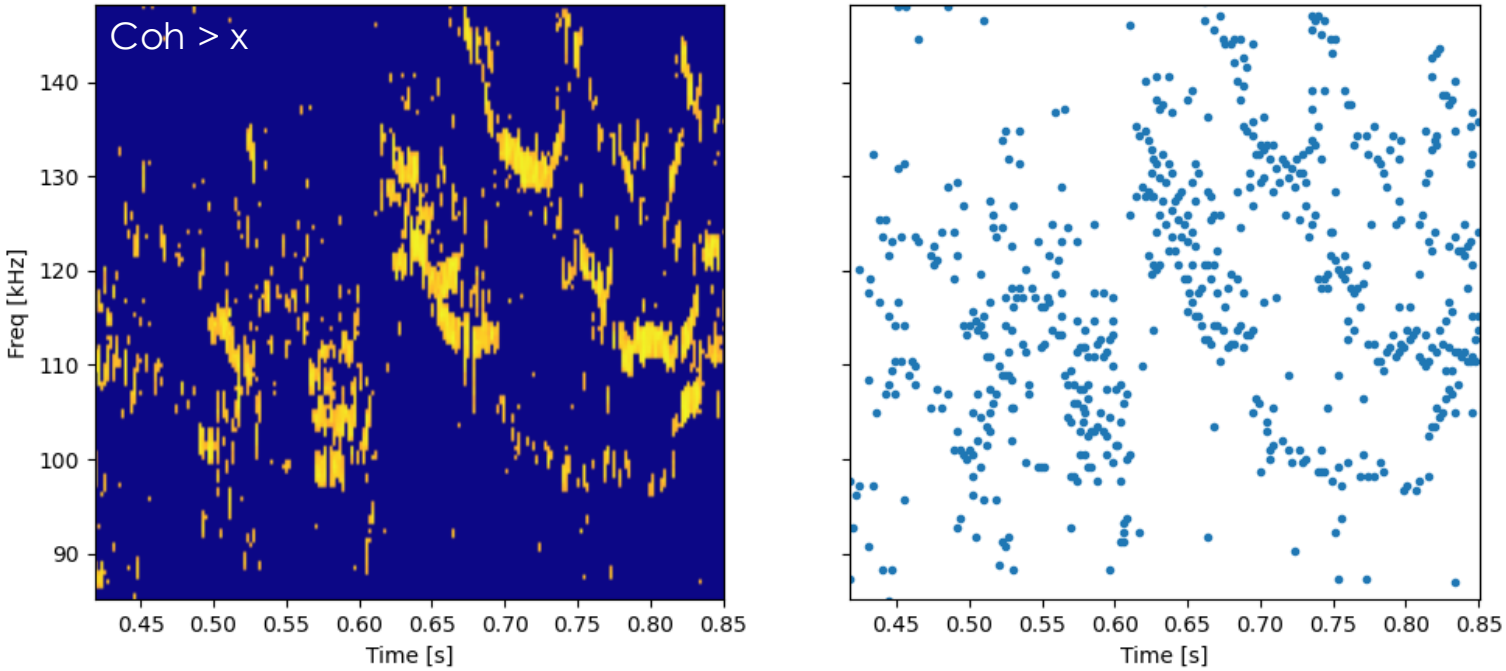
- A fast filter to find modes
- **A *Travelling Salesman* Algorithm to track modes**

The travelling salesman problem

- *"Given a list of cities and the distances between each pair of cities, what is the shortest possible route that visits each city exactly once and returns to the origin city?"*
- This is a **NP-hard** problem
- This is maybe the most intensively studied problem in optimization



Coherence matrix points larger than a threshold can be interpreted as cities to visit

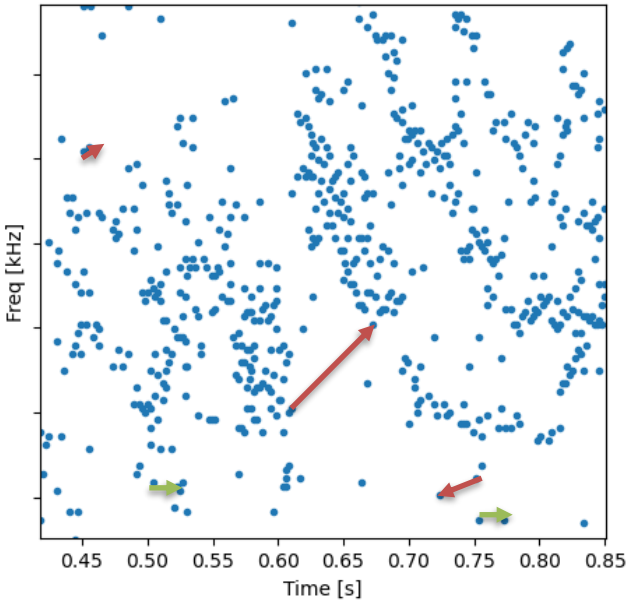


Number of trucks = number of modes

Truck route = mode f(t)

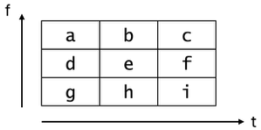
Particular *'traffic rules'* makes our problem afforable

- Roads are only going east-> A mode must have only one frequency at each time
- Trucks must stop after only a few miles -> There can't be large $\delta t, \delta f$
- Trucks must do a minimum number of stops -> modes must last a minimum of x ms

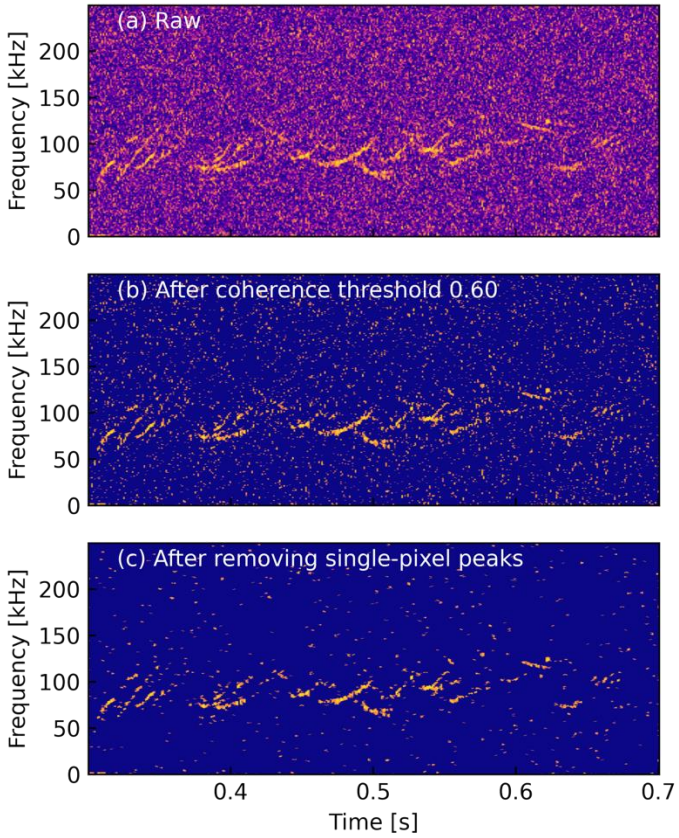


Basic filters are applied prior to run the salesman calculation

- 1st: Threshold in the coherence
- 2nd: Delete orphan nodes

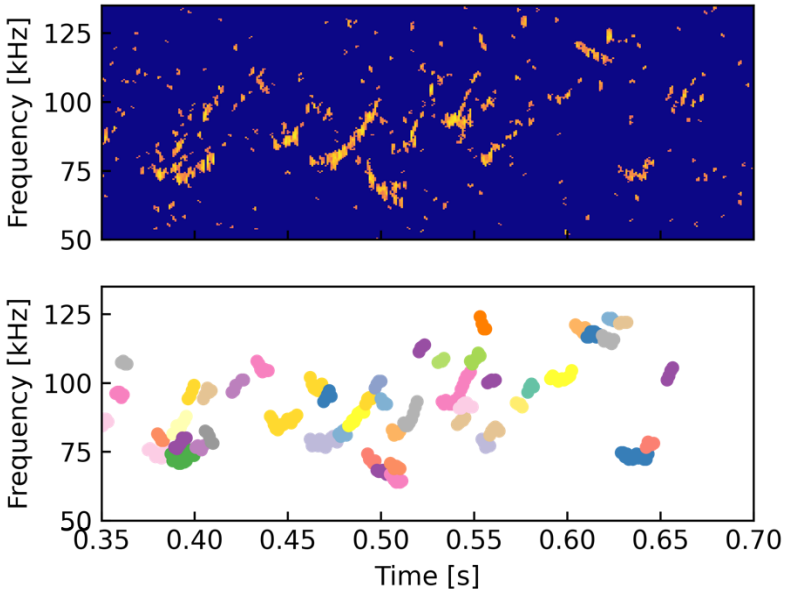


- Further methods can be applied:
  - Phase filter to mitigate thermal noise
  - Cross-power threshold to eliminate spurious fluctuations
  - …



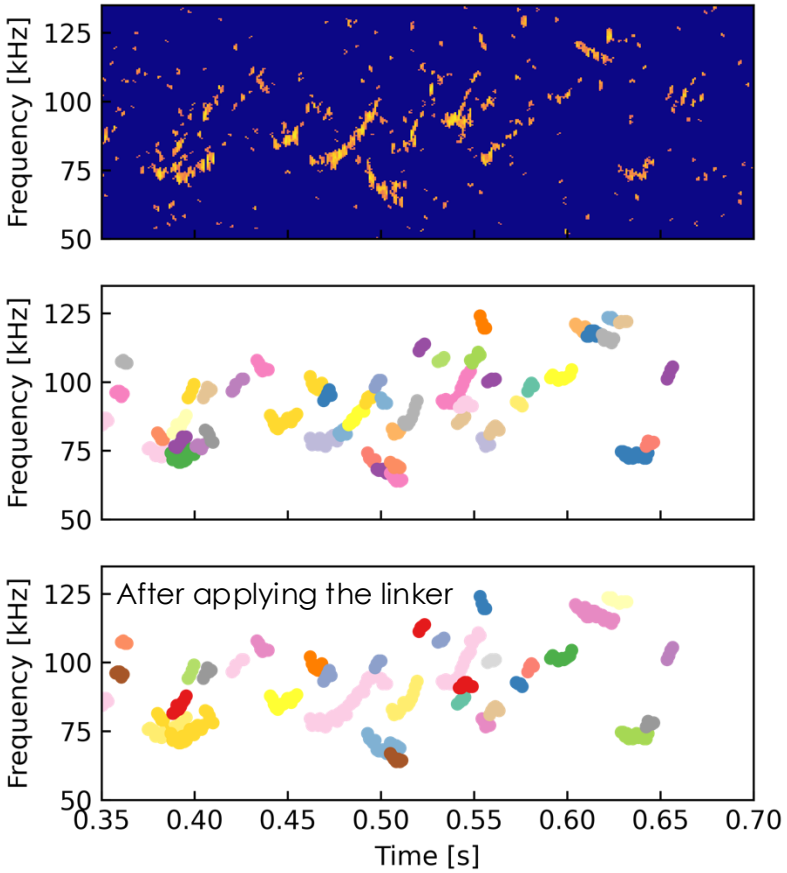
Cost functions allows us to determine the paths

- Simple cost function suffices:
  - $C(\delta f, \delta t) = \alpha|\delta t|^n + \beta|\delta f|^m$
- Others physics informed terms could be added:
  - Change in the gradient of f(t)
  - For TAEs: If density is provided, penalize changes in f(t) not following changes in ne(t)
  - For RSAEs: Penalize $\delta f < 0$
  - If n-phasogram is available: penalize changes in mode number
  - …



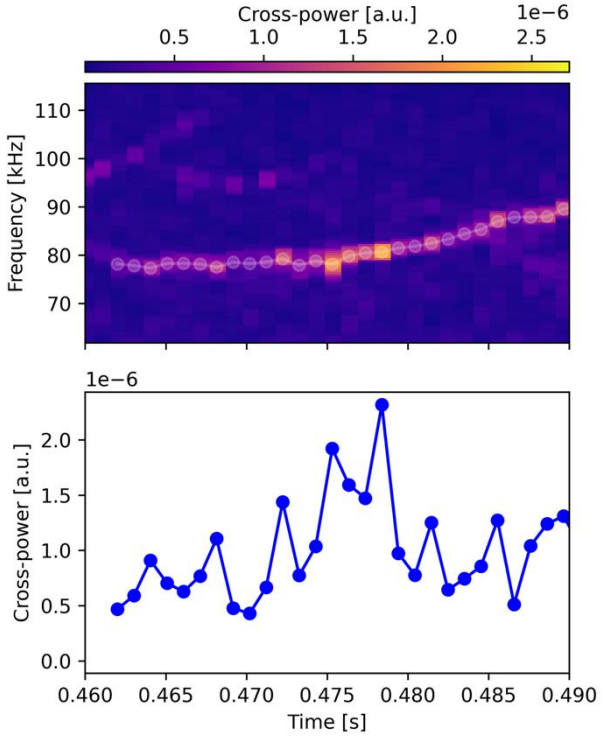
Mode linker remove spurious sub-tracks and solve crossing

- Simple cost function also suffices:
  - $C(\delta f, \delta t) = \alpha|\delta t|^n + \beta|\delta f|^m + \gamma\delta\frac{df}{dt}$
  - Limited to maximums $\delta f, \delta t, \delta\frac{df}{dt}$



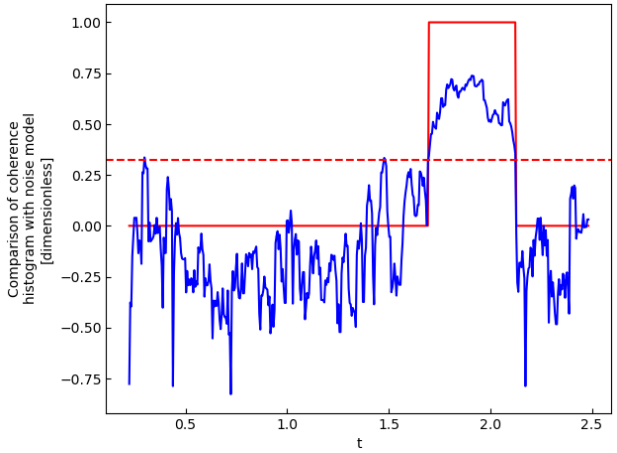
Gaussian fits allows us to get mode amplitude and refine location

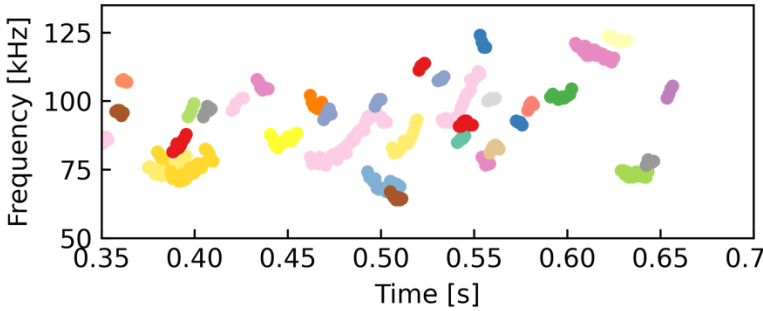
- Each time slice is fit to a multi-gaussian
  - Initial guesses for parameters taken from the mode tracks
- Helps to further refine the track f(t)



Conclusions

- A simple comparison with a polynomial fit suffices to detect times where modes are present
- Mode tracking can be written as a salesman problem
- Cost function allows us to introduce all our physical knowledge
- Code is available in ep-toolbox (we need to clean this!)





Simple polynomial fit suffice to find the modes

- Collect the coherence early in the discharge where no coherent mode is expected
- Fit to get the model parameters
- Compare with noise times to get confidence intervals
- Compare with experiment time slices to get acceptance
  - $Accept = \frac{\int H(t_i) - H_{fit} dx}{\int H_{fit} dx} > \mu + k\sigma$

